

# DAC Infrastructure Intelligence Watcher - Custom Range Report

Range: 30d

Report layer version: v1.8.0

Generated at UTC: 2026-06-29T04:57:33.921200+00:00

This portrait PDF is generated from the structured custom range JSON summary.

## 1. Range Summary

| Field                 | Value                                         |
|-----------------------|-----------------------------------------------|
| Project               | DAC Infrastructure Intelligence Watcher       |
| Snapshot count        | 78                                            |
| First snapshot        | 2026-06-04T09:54:55-772035+00-00-health.json  |
| Latest snapshot       | 2026-06-28T14:43:06-952490+00-00-health.json  |
| First checked at UTC  | 2026-06-04T09:54:55.772035+00:00              |
| Latest checked at UTC | 2026-06-28T14:43:06.952490+00:00              |
| Overall status counts | DEGRADED: 32, HEALTHY: 35, PARTIAL_OUTAGE: 11 |

## 2. Endpoint Status Counts

| Endpoint             | Status counts                            |
|----------------------|------------------------------------------|
| official_public_rpc  | DEGRADED: 32, HEALTHY: 35, UNHEALTHY: 11 |
| explorer_web         | HEALTHY: 78                              |
| primary_explorer_api | HEALTHY: 78                              |

## 3. Response-Time Summary

| Endpoint             | Average    | Max        | Response class counts                       |
|----------------------|------------|------------|---------------------------------------------|
| official_public_rpc  | 6033.54 ms | 27901.0 ms | FAST: 3, MODERATE: 8, SLOW: 57, UNKNOWN: 10 |
| explorer_web         | 710.69 ms  | 2299.0 ms  | FAST: 16, MODERATE: 52, UNKNOWN: 10         |
| primary_explorer_api | 466.93 ms  | 1157.0 ms  | FAST: 38, MODERATE: 30, UNKNOWN: 10         |

## 4. Snapshot Timeline - Availability

| #  | Checked at UTC                   | Overall        | RPC       | Web     | API     |
|----|----------------------------------|----------------|-----------|---------|---------|
| 1  | 2026-06-04T09:54:55.772035+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 2  | 2026-06-04T09:59:08.192048+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 3  | 2026-06-04T09:59:48.937274+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 4  | 2026-06-04T10:00:10.486388+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 5  | 2026-06-04T10:02:58.023788+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 6  | 2026-06-04T11:35:39.447567+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 7  | 2026-06-04T11:36:10.013189+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 8  | 2026-06-04T11:38:16.082796+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 9  | 2026-06-04T11:38:26.409385+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 10 | 2026-06-04T11:50:50.288397+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 11 | 2026-06-04T16:59:50.793517+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 12 | 2026-06-04T17:34:38.689983+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 13 | 2026-06-04T17:39:38.929050+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 14 | 2026-06-04T19:47:12.825324+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 15 | 2026-06-06T12:16:24.339927+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 16 | 2026-06-06T15:41:44.603473+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 17 | 2026-06-07T08:29:35.513958+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 18 | 2026-06-07T10:41:05.109781+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 19 | 2026-06-07T12:12:00.907268+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 20 | 2026-06-07T17:06:43.581492+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 21 | 2026-06-08T01:59:16.847351+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 22 | 2026-06-08T06:47:13.271825+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 23 | 2026-06-09T01:30:52.519172+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 24 | 2026-06-09T05:51:55.728730+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 25 | 2026-06-09T12:21:12.097443+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 26 | 2026-06-09T15:36:15.483799+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 27 | 2026-06-09T17:35:46.336066+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 28 | 2026-06-09T21:23:45.229029+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 29 | 2026-06-10T12:00:49.579476+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 30 | 2026-06-10T16:06:33.205459+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 31 | 2026-06-10T18:40:39.968153+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 32 | 2026-06-10T21:13:30.728841+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 33 | 2026-06-10T22:57:26.492242+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 34 | 2026-06-11T00:13:29.727818+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 35 | 2026-06-13T10:43:09.685105+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 36 | 2026-06-13T12:15:24.292365+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 37 | 2026-06-13T17:43:45.241764+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 38 | 2026-06-13T18:57:21.166839+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 39 | 2026-06-14T02:47:39.011114+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 40 | 2026-06-14T07:17:43.721261+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 41 | 2026-06-14T15:06:02.854660+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 42 | 2026-06-14T17:46:11.842614+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |

| #  | Checked at UTC                   | Overall        | RPC       | Web     | API     |
|----|----------------------------------|----------------|-----------|---------|---------|
| 43 | 2026-06-15T16:32:07.477470+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 44 | 2026-06-15T20:07:32.425470+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 45 | 2026-06-18T11:35:52.595819+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 46 | 2026-06-18T14:34:01.163762+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 47 | 2026-06-18T17:47:21.319826+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 48 | 2026-06-19T00:07:49.114576+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 49 | 2026-06-19T05:22:45.224068+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 50 | 2026-06-19T13:33:33.772138+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 51 | 2026-06-19T16:20:47.830658+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 52 | 2026-06-21T07:05:49.768754+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 53 | 2026-06-21T10:45:15.026750+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 54 | 2026-06-22T16:01:14.927486+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 55 | 2026-06-22T21:50:02.920913+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 56 | 2026-06-23T05:34:47.341574+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 57 | 2026-06-23T08:46:04.761597+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 58 | 2026-06-24T16:00:28.478127+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 59 | 2026-06-24T18:06:14.278129+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 60 | 2026-06-25T07:53:39.662511+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 61 | 2026-06-25T10:36:00.952733+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 62 | 2026-06-25T15:44:39.846328+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 63 | 2026-06-25T19:29:25.516852+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 64 | 2026-06-26T08:18:53.347052+00:00 | PARTIAL_OUTAGE | UNHEALTHY | HEALTHY | HEALTHY |
| 65 | 2026-06-26T11:20:30.135425+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 66 | 2026-06-26T20:55:49.405430+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 67 | 2026-06-26T22:06:06.469588+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 68 | 2026-06-26T23:22:07.890400+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 69 | 2026-06-27T13:14:03.249844+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 70 | 2026-06-27T14:55:49.430298+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 71 | 2026-06-27T16:00:02.127479+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 72 | 2026-06-27T17:16:45.438386+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 73 | 2026-06-27T19:57:07.933682+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 74 | 2026-06-27T20:59:11.427839+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 75 | 2026-06-28T06:01:40.043636+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 76 | 2026-06-28T09:30:32.330521+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |
| 77 | 2026-06-28T12:51:16.533873+00:00 | DEGRADED       | DEGRADED  | HEALTHY | HEALTHY |
| 78 | 2026-06-28T14:43:06.952490+00:00 | HEALTHY        | HEALTHY   | HEALTHY | HEALTHY |

## 5. Snapshot Timeline - Response Classes

| # | Checked at UTC                   | RPC class | Web class | API class | RPC block |
|---|----------------------------------|-----------|-----------|-----------|-----------|
| 1 | 2026-06-04T09:54:55.772035+00:00 | N/A       | N/A       | N/A       | N/A       |
| 2 | 2026-06-04T09:59:08.192048+00:00 | N/A       | N/A       | N/A       | 14952932  |
| 3 | 2026-06-04T09:59:48.937274+00:00 | N/A       | N/A       | N/A       | 14952932  |
| 4 | 2026-06-04T10:00:10.486388+00:00 | N/A       | N/A       | N/A       | N/A       |

| #  | Checked at UTC                   | RPC class | Web class | API class | RPC block |
|----|----------------------------------|-----------|-----------|-----------|-----------|
| 5  | 2026-06-04T10:02:58.023788+00:00 | N/A       | N/A       | N/A       | 14952932  |
| 6  | 2026-06-04T11:35:39.447567+00:00 | N/A       | N/A       | N/A       | 14953437  |
| 7  | 2026-06-04T11:36:10.013189+00:00 | N/A       | N/A       | N/A       | 14953447  |
| 8  | 2026-06-04T11:38:16.082796+00:00 | N/A       | N/A       | N/A       | N/A       |
| 9  | 2026-06-04T11:38:26.409385+00:00 | N/A       | N/A       | N/A       | N/A       |
| 10 | 2026-06-04T11:50:50.288397+00:00 | N/A       | N/A       | N/A       | 14913037  |
| 11 | 2026-06-04T16:59:50.793517+00:00 | MODERATE  | MODERATE  | MODERATE  | N/A       |
| 12 | 2026-06-04T17:34:38.689983+00:00 | SLOW      | MODERATE  | MODERATE  | 14954882  |
| 13 | 2026-06-04T17:39:38.929050+00:00 | SLOW      | MODERATE  | FAST      | N/A       |
| 14 | 2026-06-04T19:47:12.825324+00:00 | MODERATE  | MODERATE  | MODERATE  | 14955409  |
| 15 | 2026-06-06T12:16:24.339927+00:00 | SLOW      | MODERATE  | FAST      | 14961923  |
| 16 | 2026-06-06T15:41:44.603473+00:00 | MODERATE  | MODERATE  | FAST      | 14962325  |
| 17 | 2026-06-07T08:29:35.513958+00:00 | SLOW      | MODERATE  | FAST      | N/A       |
| 18 | 2026-06-07T10:41:05.109781+00:00 | SLOW      | MODERATE  | FAST      | 14961068  |
| 19 | 2026-06-07T12:12:00.907268+00:00 | SLOW      | MODERATE  | MODERATE  | N/A       |
| 20 | 2026-06-07T17:06:43.581492+00:00 | SLOW      | FAST      | FAST      | 14972051  |
| 21 | 2026-06-08T01:59:16.847351+00:00 | SLOW      | FAST      | FAST      | N/A       |
| 22 | 2026-06-08T06:47:13.271825+00:00 | SLOW      | FAST      | FAST      | 14975326  |
| 23 | 2026-06-09T01:30:52.519172+00:00 | SLOW      | MODERATE  | FAST      | N/A       |
| 24 | 2026-06-09T05:51:55.728730+00:00 | SLOW      | MODERATE  | MODERATE  | 14980864  |
| 25 | 2026-06-09T12:21:12.097443+00:00 | SLOW      | MODERATE  | FAST      | 14982303  |
| 26 | 2026-06-09T15:36:15.483799+00:00 | SLOW      | MODERATE  | MODERATE  | 14983164  |
| 27 | 2026-06-09T17:35:46.336066+00:00 | SLOW      | MODERATE  | FAST      | 14983257  |
| 28 | 2026-06-09T21:23:45.229029+00:00 | SLOW      | FAST      | FAST      | 14984591  |
| 29 | 2026-06-10T12:00:49.579476+00:00 | SLOW      | FAST      | FAST      | N/A       |
| 30 | 2026-06-10T16:06:33.205459+00:00 | SLOW      | FAST      | FAST      | 14988914  |
| 31 | 2026-06-10T18:40:39.968153+00:00 | SLOW      | MODERATE  | FAST      | N/A       |
| 32 | 2026-06-10T21:13:30.728841+00:00 | SLOW      | FAST      | FAST      | 14990315  |
| 33 | 2026-06-10T22:57:26.492242+00:00 | SLOW      | FAST      | FAST      | 14990731  |
| 34 | 2026-06-11T00:13:29.727818+00:00 | SLOW      | MODERATE  | FAST      | 14991035  |
| 35 | 2026-06-13T10:43:09.685105+00:00 | SLOW      | MODERATE  | FAST      | 15005070  |
| 36 | 2026-06-13T12:15:24.292365+00:00 | MODERATE  | MODERATE  | FAST      | 15005442  |
| 37 | 2026-06-13T17:43:45.241764+00:00 | FAST      | MODERATE  | MODERATE  | N/A       |
| 38 | 2026-06-13T18:57:21.166839+00:00 | SLOW      | MODERATE  | MODERATE  | 14933447  |
| 39 | 2026-06-14T02:47:39.011114+00:00 | FAST      | MODERATE  | FAST      | N/A       |
| 40 | 2026-06-14T07:17:43.721261+00:00 | SLOW      | MODERATE  | MODERATE  | 15007750  |
| 41 | 2026-06-14T15:06:02.854660+00:00 | SLOW      | MODERATE  | MODERATE  | 14933447  |
| 42 | 2026-06-14T17:46:11.842614+00:00 | SLOW      | MODERATE  | MODERATE  | 15012444  |
| 43 | 2026-06-15T16:32:07.477470+00:00 | MODERATE  | MODERATE  | MODERATE  | N/A       |
| 44 | 2026-06-15T20:07:32.425470+00:00 | MODERATE  | FAST      | FAST      | 15018851  |
| 45 | 2026-06-18T11:35:52.595819+00:00 | SLOW      | FAST      | FAST      | N/A       |
| 46 | 2026-06-18T14:34:01.163762+00:00 | SLOW      | MODERATE  | MODERATE  | 15034798  |
| 47 | 2026-06-18T17:47:21.319826+00:00 | SLOW      | MODERATE  | FAST      | 15035570  |
| 48 | 2026-06-19T00:07:49.114576+00:00 | SLOW      | FAST      | FAST      | 15035796  |

| #  | Checked at UTC                   | RPC class | Web class | API class | RPC block |
|----|----------------------------------|-----------|-----------|-----------|-----------|
| 49 | 2026-06-19T05:22:45.224068+00:00 | MODERATE  | MODERATE  | FAST      | 15038219  |
| 50 | 2026-06-19T13:33:33.772138+00:00 | FAST      | MODERATE  | MODERATE  | 15040313  |
| 51 | 2026-06-19T16:20:47.830658+00:00 | MODERATE  | FAST      | FAST      | 15040982  |
| 52 | 2026-06-21T07:05:49.768754+00:00 | SLOW      | MODERATE  | MODERATE  | 15050283  |
| 53 | 2026-06-21T10:45:15.026750+00:00 | SLOW      | MODERATE  | MODERATE  | 15051129  |
| 54 | 2026-06-22T16:01:14.927486+00:00 | SLOW      | MODERATE  | MODERATE  | N/A       |
| 55 | 2026-06-22T21:50:02.920913+00:00 | SLOW      | MODERATE  | MODERATE  | 15058413  |
| 56 | 2026-06-23T05:34:47.341574+00:00 | SLOW      | MODERATE  | MODERATE  | 15058413  |
| 57 | 2026-06-23T08:46:04.761597+00:00 | SLOW      | FAST      | FAST      | 15062123  |
| 58 | 2026-06-24T16:00:28.478127+00:00 | SLOW      | FAST      | FAST      | 15069578  |
| 59 | 2026-06-24T18:06:14.278129+00:00 | SLOW      | FAST      | FAST      | 15070080  |
| 60 | 2026-06-25T07:53:39.662511+00:00 | SLOW      | MODERATE  | FAST      | 15073391  |
| 61 | 2026-06-25T10:36:00.952733+00:00 | SLOW      | MODERATE  | MODERATE  | N/A       |
| 62 | 2026-06-25T15:44:39.846328+00:00 | SLOW      | MODERATE  | FAST      | 15075276  |
| 63 | 2026-06-25T19:29:25.516852+00:00 | SLOW      | MODERATE  | MODERATE  | 15076106  |
| 64 | 2026-06-26T08:18:53.347052+00:00 | SLOW      | MODERATE  | MODERATE  | N/A       |
| 65 | 2026-06-26T11:20:30.135425+00:00 | SLOW      | FAST      | FAST      | N/A       |
| 66 | 2026-06-26T20:55:49.405430+00:00 | SLOW      | MODERATE  | FAST      | 15082275  |
| 67 | 2026-06-26T22:06:06.469588+00:00 | SLOW      | MODERATE  | MODERATE  | 15082556  |
| 68 | 2026-06-26T23:22:07.890400+00:00 | SLOW      | MODERATE  | MODERATE  | 15082859  |
| 69 | 2026-06-27T13:14:03.249844+00:00 | SLOW      | MODERATE  | FAST      | N/A       |
| 70 | 2026-06-27T14:55:49.430298+00:00 | SLOW      | MODERATE  | MODERATE  | 15086595  |
| 71 | 2026-06-27T16:00:02.127479+00:00 | SLOW      | MODERATE  | FAST      | N/A       |
| 72 | 2026-06-27T17:16:45.438386+00:00 | SLOW      | MODERATE  | FAST      | 15087158  |
| 73 | 2026-06-27T19:57:07.933682+00:00 | SLOW      | MODERATE  | MODERATE  | 15087801  |
| 74 | 2026-06-27T20:59:11.427839+00:00 | SLOW      | MODERATE  | MODERATE  | 15088048  |
| 75 | 2026-06-28T06:01:40.043636+00:00 | SLOW      | MODERATE  | MODERATE  | N/A       |
| 76 | 2026-06-28T09:30:32.330521+00:00 | SLOW      | MODERATE  | MODERATE  | 15091053  |
| 77 | 2026-06-28T12:51:16.533873+00:00 | SLOW      | MODERATE  | FAST      | 15091857  |
| 78 | 2026-06-28T14:43:06.952490+00:00 | SLOW      | MODERATE  | MODERATE  | 15092303  |

## 6. Interpretation Guide

- Availability status describes whether a service is reachable and usable.
- Response class describes response-time behavior, not availability.
- Older snapshots may show UNKNOWN or null response class because response-time classification was added after the initial watcher release.
- This custom range report is independent observation material and not an official DAC service status page.

## Status & Response-Class Glossary

| Status / Class | Meaning                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| HEALTHY        | The endpoint or overall infrastructure state is reachable and behaving as expected.                               |
| DEGRADED       | The endpoint is reachable, but one or more checks or response-time indicators show reduced quality.               |
| PARTIAL_OUTAGE | At least one monitored endpoint is unavailable or failing while other endpoints remain reachable.                 |
| UNHEALTHY      | The endpoint failed required checks or did not provide usable responses.                                          |
| FAST           | The observed response-time class is fast for this watcher context.                                                |
| MODERATE       | The observed response-time class is acceptable but not fast.                                                      |
| SLOW           | The observed response-time class is slow and may indicate degraded user experience.                               |
| UNKNOWN        | The watcher could not classify the response-time state, often because older snapshots did not include this field. |
| N/A            | Not available or not applicable for the selected observation, endpoint, or historical snapshot.                   |

Prepared by **JERUZZALEM - DAC Infra Tester**.